

AP CSP | Lesson 16: Study Goal Tracker

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

12 original questions, 30 teacher marks, about 45 minutes. Q5 requires exactly TWO choices. Show predictions, traces and reasons. Braces mark blocks; each statement is on its own line.

1. What is the value of $17 \text{ MOD } 5 + 2 * 4$? [1 mark]

- A. 16 B. 10 C. 12 D. 11.4

Answer: _____

2. When $x = 4$, which expression is true? [1 mark]

- A. $(x < 4) \text{ AND } (x > 0)$ B. $\text{NOT}(x = 4)$
C. $(x = 4) \text{ AND NOT}(x > 5)$ D. $(x > 5) \text{ OR } (x < 0)$

Answer: _____

3. IF(score $\geq$ 50) displays Pass; ELSE displays Retry. What appears when score = 50? [1 mark]

- A. Retry B. Both messages C. Neither message D. Pass

Answer: _____

4. Start total = 1. Repeat total $\leftarrow$ total * 2 three times. What is total? [1 mark]

- A. 8 B. 6 C. 4 D. 16

Answer: _____

5. Which TWO statements about AP REPEAT UNTIL are correct? [1 mark]

- A. It stops when the condition is false. B. An initially true condition can give zero body runs.
C. Any changing variable guarantees it ends. D. A condition that never becomes true can cause an infinite loop.

Answer: _____

6. What is $P(\text{RANDOM}(1, 5) \leq 2)$? [1 mark]

- A. 1/2 B. 1/5 C. 2/5 D. 3/5

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

```
x ← 3
y ← x + 2
x ← y * 2
y ← x MOD 4
```

- (a) Give y after line 2. (b) Give x after line 3. (c) Give final y.
(d) Evaluate `CONCAT("AP", " CSP")`. `CONCAT` joins strings without adding spaces.

8. [4 marks]

Rule: if age ≥ 12 , check member. If member is true, display Member; otherwise display Guest. If age < 12 , display Junior.

- (a) Predict age 12/member false. (b) Predict age 11/member true.
(c) Predict age 12/member true. (d) Explain why member need not be checked for age 11.

9. [4 marks]

```
n ← 2
REPEAT UNTIL(n ≥ 7)
{
  n ← n + 2
}
```

- (a) Give final n. (b) Give body count. (c) Would replacing $\geq$ with $=$ terminate? Explain. (d) Give a start value making the original loop run zero times.

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Written responses | Questions 10-12

10. [4 marks]

A fair independent coin model uses `RANDOM(1, 2)`, with 1 = Heads. It gives 8 Heads in 20 trials.

(a) Give theoretical $P(\text{Heads})$. (b) Calculate observed frequency.

(c) Must the next trial be Heads? Explain. (d) State one reason a real coin experiment could differ from this simplified model.

11. [4 marks]

Study Goal Tracker reads three valid daily values and counts minutes ≥ 30 . After three days it displays count, then On track if count ≥ 2 , else Try again.

(a) For 29, 30, 31, give count and final message.

(b) For 0, 0, 0, give count and final message.

(c) For 30, 30, 30, give count and final message.

(d) Explain why 29 and 30 form a useful boundary pair.

12. [4 marks]

A faulty tracker resets `count ← 0` inside its three-day loop. All inputs are 30.

(a) Give the faulty final count. (b) Write the corrected initialization and loop outline.

(c) Give the corrected final count and final message.

(d) Write a revision record linking this change to a test.
